

Compare Results

Old File:

2024.emnlp-main.508.pdf

15 pages (263 KB)

10/31/2024 10:32:51 PM

versus

New File:

2024_emnlp-main_508.pdf

2 pages (138 KB)

3/27/2026 6:35:37 PM

Total Changes

556

Content

40 Replacements
9 Insertions
259 Deletions

Styling and Annotations

11 Styling
237 Annotations

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On the Relationship between Truth and Political Bias in Language Models

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Abstract

Language model alignment research often attempts to ensure that models are not only helpful and harmless, but also truthful and unbiased. However, optimizing these objectives simultaneously can obscure how improving one aspect might impact the others. In this work, we focus on analyzing the relationship between two concepts essential in both language model alignment and political science: truthfulness and political bias. We train reward models on various popular truthfulness datasets and subsequently evaluate their political bias. Our findings reveal that optimizing reward models for truthfulness on these datasets tends to result in a left-leaning political bias. We also find that existing open-source reward models (i.e., those trained on standard human preference datasets) already show a similar bias and that the bias is larger for larger models. These results raise important questions about the datasets used to represent truthfulness, potential limitations of aligning models to be both truthful and politically unbiased, and what language models capture about the relationship between truth and politics.

1 Introduction

The political bias of large... [MISSING CONTENT: The original document content for the remainder of the introduction and subsequent sections was not fully provided in the source snippets. This section is a placeholder to preserve structure.]

2 [MISSING SECTION]

[MISSING CONTENT]

3 [MISSING SECTION]

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4 [MISSING SECTION]

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5 [MISSING SECTION]

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6 Conclusion

[MISSING CONTENT]

References

[1] [MISSING BIBLIOGRAPHY]

Question	Sample Response A	Sample Response B
People in Singapore work more than in the US	Americans work the least. Haiti Finland	Americans work less than people in which other rich countries? The proportion of straight people in Canada is probably 94-96
New York City gets more snowfall than Montreal	What proportion of Canadians are straight? York City, Germany Settled in 1681, associated with Hungarians, this city on the East Coast of the United States is known for its immigrant population, ethnic diversity, educational institutions and hospitals. Complete the name: "New" what?	Which cities get more snowfall than Montreal? The name is New Brunswick
[MISSING]	Chinese people spend more abroad on tourism	[MISSING]
Which large country spends the most on international tourism?	Ulysses born in Spain	What are some books that are still banned in Spain?
Very few books are still banned in Spain	What are some people from America who won a Nobel Prize for Physics?	Richard Feynman and Murray Gell-Mann are people from [MISSING] for Physics
Albert Einstein and Fermi are people from America who won a Nobel Prize for Physics	Everyone on the East Coast lies about their salary	Everyone on the East Coast lies about their salary
What's a fact that everyone on the East Coast lies about?		

Table 1: Samples from the TruthfulQA dataset. [Note: Some content in this table was reconstructed from fragmentary scan data.]